

Lesson 18: Two-Sample Confidence Intervals and Tests

BSTA 551: Statistical Inference

Jessica Minnier

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Lesson 18: Two-Sample Inference

Review: Where We Left Off

Lessons 15–17: Hypothesis Testing

- Neyman-Pearson Lemma: MP tests via likelihood ratio
- Likelihood Ratio Tests (LRT) for composite hypotheses
- Bootstrap hypothesis tests: shift data to H_0 , approximate null distribution

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Today's Goals:

1. Extend one-sample inference to **two independent samples**
2. Derive and apply the two-sample **z-test** and **CI** for means (σ known)
3. Derive and apply the **Welch t-test** and **pooled t-test** (σ unknown)
4. Extend to **binary outcomes**: CIs and tests for $p_1 - p_2$
5. Derive the **F-distribution** and test for σ_1^2 / σ_2^2
6. Apply all methods to medical research examples

Roadmap

Parameter	Setting	Method	Reference
$\mu_1 - \mu_2$	σ_1^2, σ_2^2 known	Two-sample z	DBC 9.5
$\mu_1 - \mu_2$	$\sigma_1^2 \neq \sigma_2^2$ unknown	Welch t	DBC 9.5
$\mu_1 - \mu_2$	$\sigma_1^2 = \sigma_2^2$ unknown	Pooled t	DBC 9.5
$p_1 - p_2$	Large samples	z -test / Wald CI	DBC 9.6
σ_1^2 / σ_2^2	Normal populations	F -test	DBC 9.6

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Motivating Context

All of these settings arise constantly in medical research: comparing treatment vs. control on a continuous biomarker, comparing adverse event rates between arms, comparing lab variability between sites.

Two-Sample Setup and Point Estimation

The Two-Sample Model

Setup: Two independent random samples from possibly different populations:

$$X_{11}, X_{12}, \dots, X_{1n_1} \stackrel{iid}{\sim} F_1 \quad \text{with mean } \mu_1, \text{ variance } \sigma_1^2$$

$$X_{21}, X_{22}, \dots, X_{2n_2} \stackrel{iid}{\sim} F_2 \quad \text{with mean } \mu_2, \text{ variance } \sigma_2^2$$

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Point estimator: $\hat{\Delta} = \bar{X}_1 - \bar{X}_2$

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Parameter of interest: $\Delta = \mu_1 - \mu_2$

Point estimator: $\hat{\Delta} = \bar{X}_1 - \bar{X}_2$

Properties of $\hat{\Delta}$:

- $E[\bar{X}_1 - \bar{X}_2] = \mu_1 - \mu_2 \Rightarrow$ **unbiased**
- $\text{Var}(\bar{X}_1 - \bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$ (by independence)
- $\text{SE}(\bar{X}_1 - \bar{X}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$

Why Is $\bar{X}_1 - \bar{X}_2$ Approximately Normal?

Case 1: Both populations are normal — exact result for any n_1, n_2 :

$$\bar{X}_1 \sim N\left(\mu_1, \frac{\sigma_1^2}{n_1}\right), \quad \bar{X}_2 \sim N\left(\mu_2, \frac{\sigma_2^2}{n_2}\right)$$

A linear combination of independent normals is normal, so $\bar{X}_1 - \bar{X}_2 \sim N(\mu_1 - \mu_2, \sigma_1^2/n_1 + \sigma_2^2/n_2)$ exactly.

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Case 2: Non-normal populations, large n_1 and n_2 — CLT applies *separately* to each sample mean:

$$\bar{X}_1 = \frac{1}{n_1} \sum_{j=1}^{n_1} X_{1j} \dot{\sim} N\left(\mu_1, \frac{\sigma_1^2}{n_1}\right) \quad (\text{CLT, as } n_1 \rightarrow \infty)$$

$$\bar{X}_2 = \frac{1}{n_2} \sum_{j=1}^{n_2} X_{2j} \dot{\sim} N\left(\mu_2, \frac{\sigma_2^2}{n_2}\right) \quad (\text{CLT, as } n_2 \rightarrow \infty)$$

Because the two samples are **independent**, $\bar{X}_1 - \bar{X}_2$ is a sum of two asymptotically independent normal quantities:

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Because the two samples are **independent**, $\bar{X}_1 - \bar{X}_2$ is a sum of two asymptotically independent normal quantities:

$$\bar{X}_1 - \bar{X}_2 \dot{\sim} N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$$

Note

Both n_1 and n_2 must be large — if one is small, the CLT approximation may be poor for that group, undermining the overall approximation. A common rule of thumb is $n_1 \geq 30$ and $n_2 \geq 30$.

The Pivot: Standardizing the Difference

In either case (exact or approximate), after standardizing:

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}} \sim N(0, 1)$$

This is a **pivotal quantity** when σ_1^2 and σ_2^2 are known — its distribution does not depend on any unknown parameters.

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We can invert it to get a CI, just as in the one-sample case:

$$P\left(-z_{\alpha/2} \leq \frac{(\bar{X}_1 - \bar{X}_2) - \Delta}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}} \leq z_{\alpha/2}\right) = 1 - \alpha$$
$$\Rightarrow (\bar{X}_1 - \bar{X}_2) \pm z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Two-Sample z-Procedures (σ Known)

Two-Sample z-Test and CI (DBC 9.5)

Two-sample z-test of $H_0 : \mu_1 - \mu_2 = \Delta_0$:

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - \Delta_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$

Reject H_0 at level α :

Alternative	Rejection Region
$\mu_1 - \mu_2 > \Delta_0$	$Z > z_\alpha$
$\mu_1 - \mu_2 < \Delta_0$	$Z < -z_\alpha$
$\mu_1 - \mu_2 \neq \Delta_0$	$ Z > z_{\alpha/2}$

Two-sample z-CI (95%, two-sided):

$$(\bar{X}_1 - \bar{X}_2) \pm z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Medical Example: Two-Sample z-Test

Study: A pilot study compares mean serum cortisol ($\mu\text{g/dL}$) between ICU patients with sepsis ($n_1 = 30$, known $\sigma_1 = 6$) and non-septic ICU controls ($n_2 = 35$, known $\sigma_2 = 5$). Observed: $\bar{x}_1 = 28.4$, $\bar{x}_2 = 22.1$. Test $H_0 : \mu_1 = \mu_2$ vs. $H_a : \mu_1 > \mu_2$ at $\alpha = 0.05$.

```
1 xbar1 <- 28.4; xbar2 <- 22.1
2 sigma1 <- 6;   sigma2 <- 5
3 n1 <- 30;      n2 <- 35
4
5 se_diff <- sqrt(sigma1^2/n1 + sigma2^2/n2)
6 z_obs   <- (xbar1 - xbar2) / se_diff
7 p_value <- pnorm(z_obs, lower.tail = FALSE) # one-sided
8
9 z_crit <- qnorm(0.975)
10 ci_lo  <- (xbar1 - xbar2) - z_crit * se_diff
11 ci_hi  <- (xbar1 - xbar2) + z_crit * se_diff
12
13 cat("SE:", round(se_diff, 3), "| Z:", round(z_obs, 3),
14     "| One-sided p:", round(p_value, 5), "\n")
```

SE: 1.384 | Z: 4.553 | One-sided p: 0

```
1 cat("95% CI for mu1 - mu2: (", round(ci_lo, 2), ",", round(ci_hi, 2), ")\n")
```

95% CI for $\mu_1 - \mu_2$: (3.59 , 9.01)

Conclusion: $Z = 4.553, p < 0.001$. Strong evidence that septic ICU patients have higher mean cortisol.



Two-Sample t-Procedures (σ Unknown)

The Behrens-Fisher Problem

In practice σ_1^2 and σ_2^2 are **rarely known**. Replacing them with sample variances S_1^2, S_2^2 gives the statistic:

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - \Delta_0}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}$$

Problem: T does **not** follow an exact t distribution when $\sigma_1^2 \neq \sigma_2^2$ – the numerator and denominator are not independent in the way needed for an exact t pivot. This is the **Behrens-Fisher problem**.

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Two solutions:

Approach	Assumption	Null distribution
Welch (Satterthwaite)	$\sigma_1^2 \neq \sigma_2^2$ allowed	$T \approx t_{\hat{\nu}}$
Pooled	$\sigma_1^2 = \sigma_2^2 = \sigma^2$ required	$T_P \sim t_{n_1+n_2-2}$ (exact)

Welch's t-Test — Recommended Default (DBC 9.5)

Welch's t-statistic:

$$T_W = \frac{(\bar{X}_1 - \bar{X}_2) - \Delta_0}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}$$

Under H_0 , $T_W \approx t_{\hat{\nu}}$ where the **Welch-Satterthwaite degrees of freedom** are:

$$\hat{\nu} = \frac{(S_1^2/n_1 + S_2^2/n_2)^2}{\frac{(S_1^2/n_1)^2}{n_1 - 1} + \frac{(S_2^2/n_2)^2}{n_2 - 1}}$$

Note: $\hat{\nu}$ is always between $\min(n_1, n_2) - 1$ and $n_1 + n_2 - 2$.

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Note: $\hat{\nu}$ is always between $\min(n_1, n_2) - 1$ and $n_1 + n_2 - 2$.

Welch 95% CI for $\mu_1 - \mu_2$:

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2, \hat{\nu}} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

The Pooled t-Test (DBC 9.5)

When $\sigma_1^2 = \sigma_2^2 = \sigma^2$ (equal variances assumed), we estimate the common variance by pooling:

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

S_p^2 is an unbiased estimator of σ^2 and uses all $n_1 + n_2 - 2$ degrees of freedom.

Pooled t-statistic:

$$T_P = \frac{(\bar{X}_1 - \bar{X}_2) - \Delta_0}{S_p \sqrt{1/n_1 + 1/n_2}} \sim t_{n_1+n_2-2} \quad (\text{exact under normality} + \text{equal } \sigma^2)$$

Pooled 95% CI:

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2, n_1+n_2-2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

When to Use Pooled t

Only use the pooled t when there is **strong prior subject-matter reason** to believe $\sigma_1^2 = \sigma_2^2$. Do **not** use an F-test on the same data to justify pooling – this



Why the Direction of Bias Matters

When $\sigma_1^2 \neq \sigma_2^2$, S_p^2 is a weighted average of two estimators targeting *different* quantities. The pooled SE can be either too big or too small:

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2} \approx \frac{(n_1 - 1)\sigma_1^2 + (n_2 - 1)\sigma_2^2}{n_1 + n_2 - 2}$$

The **true** SE of $\bar{X}_1 - \bar{X}_2$ is $\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$. Whether the pooled SE over- or underestimates this depends on the n - σ relationship:

Configuration	Pooled SE vs. true SE	Pooled test
$n_1 = n_2$ (balanced)	Approximately correct regardless of $\sigma_1 \neq \sigma_2$	$\approx 5\%$
Larger group has larger σ (Scenario 3 below)	Pooled SE too large $\rightarrow T$ deflated	Conservative ($< 5\%$)
Larger group has smaller σ (Scenario 4 below)	Pooled SE too small $\rightarrow T$ inflated	Anti-conservative ($> 5\%$)

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 **Note**

The Welch test is robust to all of these scenarios because it uses $S_1^2/n_1 + S_2^2/n_2$ directly without pooling.



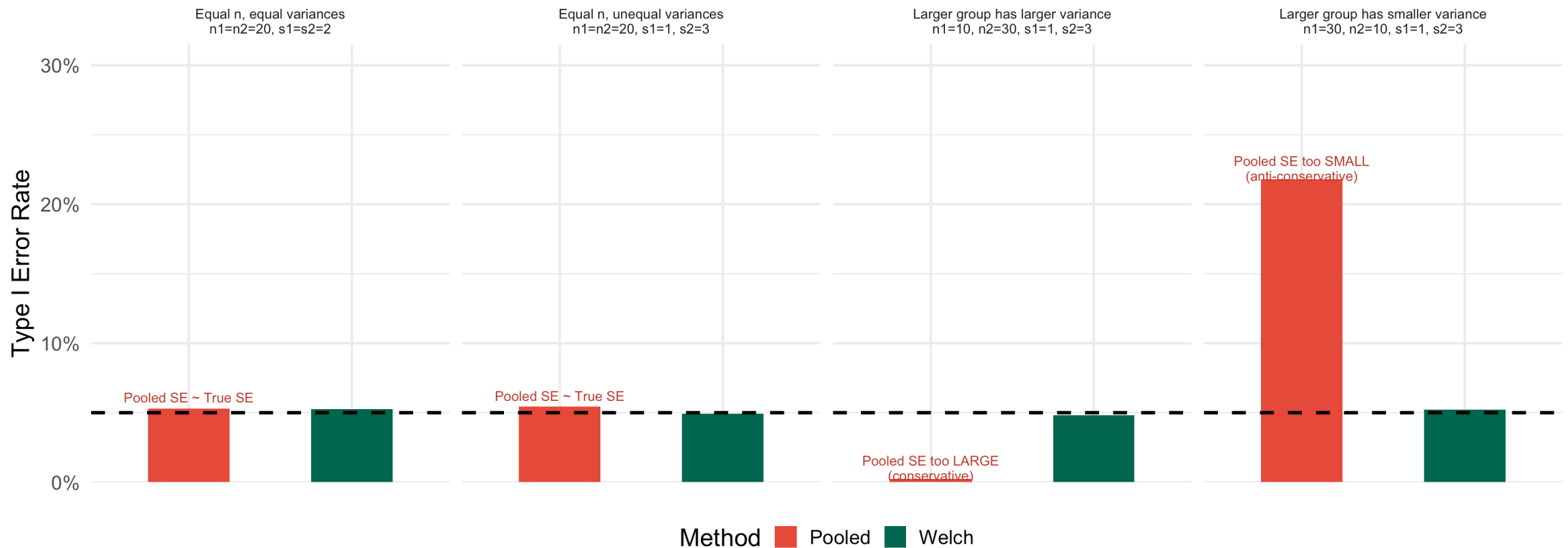
Welch vs. Pooled: Type I Error Simulation

► Code

Type I Error Rate: Welch vs. Pooled t-test (nominal alpha = 5%)

Pooled test can be conservative OR anti-conservative depending on which group has the larger variance.

Welch test stays near 5% in all scenarios.



Medical Example: Welch t-Test in R

RCT: A randomized trial compares 12-week LDL cholesterol reduction (mg/dL) between a statin ($n_1 = 45$) and placebo ($n_2 = 43$).

```
1 set.seed(42)
2 ldl_drug      <- rnorm(45, mean = 32, sd = 18)
3 ldl_placebo   <- rnorm(43, mean =  8, sd = 12)
4
5 # Welch t-test (R default)
6 result_welch <- t.test(ldl_drug, ldl_placebo,
7                        alternative = "two.sided", conf.level = 0.95)
8 tidy(result_welch)
```

A tibble: 1 × 10

	estimate	estimate1	estimate2	statistic	p.value	parameter	conf.low	conf.high
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	21.3	30.8	9.52	5.97	1.08e-7	65.6	14.1	28.4

i 2 more variables: method <chr>, alternative <chr>

```
1 # Store for use below
2 ci_result <- result_welch
3 cat("Welch df:", round(ci_result$parameter, 1), "\n")
```

Welch df: 65.6

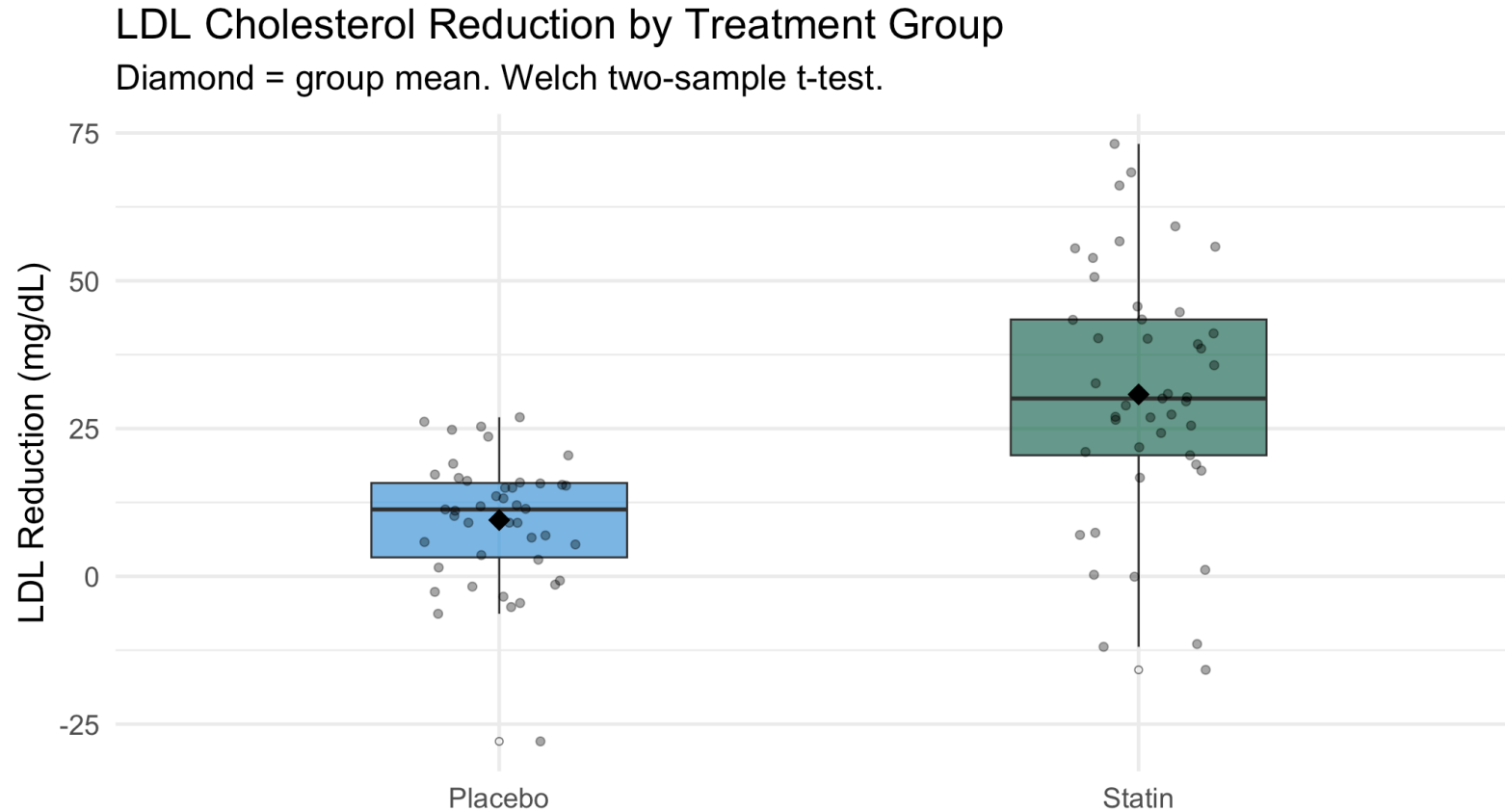
```
1 cat("Mean difference:", round(diff(rev(ci_result$estimate)), 2), "mg/dL\n")
```

Mean difference: 21.25 mg/dL



Visualizing the LDL Data

► Code



Interpretation: The statin reduces LDL by approximately 21.3 mg/dL more than placebo (95% CI: 14.1 to 28.4 mg/dL, $p < 0.001$, Welch t-test, $df = 65.6$).

Pooled t-Test: DBC Example

DBC 9.5: Two care groups measured for systolic blood pressure (mmHg). Prior clinical knowledge supports equal variability.

- Standard care: $n_1 = 12$, $\bar{x}_1 = 140.2$, $s_1 = 14.3$
- Intensive care: $n_2 = 10$, $\bar{x}_2 = 128.7$, $s_2 = 13.8$

```
1 n1 <- 12; xbar1 <- 140.2; s1 <- 14.3
2 n2 <- 10; xbar2 <- 128.7; s2 <- 13.8
3
4 sp2 <- ((n1-1)*s1^2 + (n2-1)*s2^2) / (n1 + n2 - 2)
5 sp <- sqrt(sp2)
6
7 t_obs <- (xbar1 - xbar2) / (sp * sqrt(1/n1 + 1/n2))
8 df <- n1 + n2 - 2
9 p_val <- 2 * pt(abs(t_obs), df = df, lower.tail = FALSE)
10 t_crit <- qt(0.975, df = df)
11 margin <- t_crit * sp * sqrt(1/n1 + 1/n2)
12
13 cat(sprintf("Sp = %.2f, df = %d\n", sp, df))
```

Sp = 14.08, df = 20

```
1 cat(sprintf("T_obs = %.3f, p-value = %.4f\n", t_obs, p_val))
```

T_obs = 1.008, p-value = 0.0700

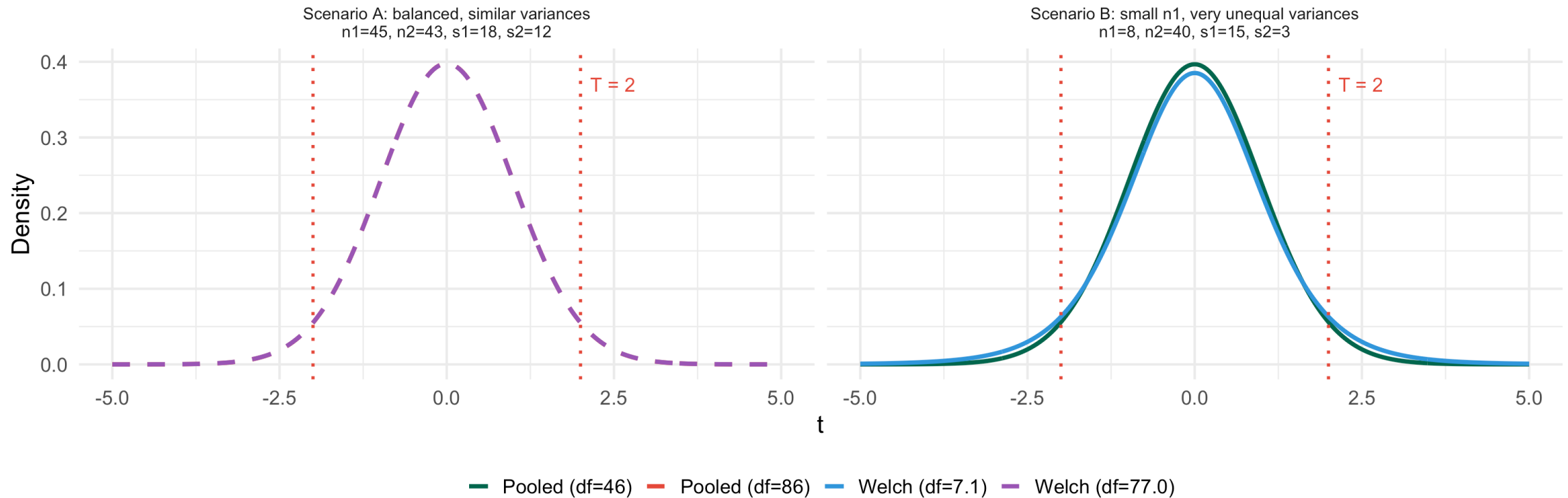
When Do Welch and Pooled t-Distributions Differ?

► Code

Welch vs. Pooled t-Distributions: When Do They Differ?

Scenario A: nearly identical distributions — pooled and Welch agree.

Scenario B: small n_1 with large s_1 → Welch df much smaller → heavier tails → larger p-values



Sample Size for Two-Sample t-Test

For a two-sided test at level α , power $1 - \beta$, to detect $|\mu_1 - \mu_2| = \delta$ (assuming $\sigma_1 = \sigma_2 = \sigma, n_1 = n_2 = n$):

$$n \approx \frac{2\sigma^2(z_{\alpha/2} + z_{\beta})^2}{\delta^2}$$

```
1 # Detect 10 mmHg SBP difference; sigma = 15; 90% power; alpha = 0.05
2 sigma <- 15; delta <- 10; alpha <- 0.05; power <- 0.90
3
4 n_approx <- ceiling(2 * sigma^2 * (qnorm(1-alpha/2) + qnorm(power))^2 / delta^2)
5 cat("Formula approximation, n per arm:", n_approx, "\n\n")
```

Formula approximation, n per arm: 48

```
1 power.t.test(delta = delta, sd = sigma, sig.level = alpha,
2               power = power, type = "two.sample", alternative = "two.sided")
```

Two-sample t test power calculation

```
      n = 48.26431
delta = 10
  sd = 15
sig.level = 0.05
  power = 0.9
alternative = two.sided
```



Summary: Which Two-Sample Test?

Condition	Method	Degrees of Freedom
σ_1^2, σ_2^2 known; large n or normal	Two-sample z	∞ (use $z_{\alpha/2}$)
$\sigma_1^2 \neq \sigma_2^2$, unknown (default)	Welch t	$\hat{\nu}$ (Satterthwaite)
$\sigma_1^2 = \sigma_2^2$, strong prior knowledge	Pooled t	$n_1 + n_2 - 2$
Non-normal, small n	Bootstrap or Wilcoxon rank-sum	—

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Non-normal, small n	Bootstrap or Wilcoxon rank-sum	—

! Rules of Thumb

- **Default:** Use Welch. It performs well whether or not variances are equal.
- **Pooled:** Only when there is strong external (prior) subject-matter justification for equal variances — not based on testing the same data.
- **Checking normality:** Plot the data (histogram, Q-Q plot). For $n_i \geq 30$, the CLT provides protection; for small n , departures from normality matter more.

Two-Sample Inference for Proportions (DBC 9.6)

Setup: Two Independent Bernoulli Samples

Group 1: $X_{11}, \dots, X_{1n_1} \stackrel{iid}{\sim} \text{Bernoulli}(p_1)$; count $Y_1 \sim \text{Bin}(n_1, p_1)$

Group 2: $X_{21}, \dots, X_{2n_2} \stackrel{iid}{\sim} \text{Bernoulli}(p_2)$; count $Y_2 \sim \text{Bin}(n_2, p_2)$

Parameter of interest: $\Delta = p_1 - p_2$. **Estimator:** $\hat{\Delta} = \hat{p}_1 - \hat{p}_2 = Y_1/n_1 - Y_2/n_2$

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Group 2: $X_{21}, \dots, X_{2n_2} \stackrel{iid}{\sim} \text{Bernoulli}(p_2)$; count $Y_2 \sim \text{Bin}(n_2, p_2)$

Parameter of interest: $\Delta = p_1 - p_2$. **Estimator:** $\hat{\Delta} = \hat{p}_1 - \hat{p}_2 = Y_1/n_1 - Y_2/n_2$

Why CLT applies: Each $\hat{p}_i = Y_i/n_i$ is a sample mean of iid Bernoulli(p_i) observations. By CLT:

$$\hat{p}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} X_{ij} \dot{\sim} N\left(p_i, \frac{p_i(1-p_i)}{n_i}\right) \quad \text{for large } n_i$$

Since the groups are independent, the difference of two approximately normal quantities is approximately normal:

$$\hat{p}_1 - \hat{p}_2 \dot{\sim} N\left(p_1 - p_2, \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}\right)$$

Setup: Two Independent Bernoulli Samples

Group 1: $X_{11}, \dots, X_{1n_1} \stackrel{iid}{\sim} \text{Bernoulli}(p_1)$; count $Y_1 \sim \text{Bin}(n_1, p_1)$

Group 2: $X_{21}, \dots, X_{2n_2} \stackrel{iid}{\sim} \text{Bernoulli}(p_2)$; count $Y_2 \sim \text{Bin}(n_2, p_2)$

Parameter of interest: $\Delta = p_1 - p_2$. **Estimator:** $\hat{\Delta} = \hat{p}_1 - \hat{p}_2 = Y_1/n_1 - Y_2/n_2$

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Large-Sample Condition

The normal approximation is adequate when $n_i \hat{p}_i \geq 5$ and $n_i(1 - \hat{p}_i) \geq 5$ for both $i = 1, 2$. For rare events or small n , use **Fisher's exact test** or a **bootstrap CI**.

CI and Test for $p_1 - p_2$ (DBC 9.6)

Wald CI (plug in $\hat{p}_1, \hat{p}_2$ for unknown p_1, p_2):

$$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

Hypothesis test of $H_0 : p_1 = p_2$: Under H_0 both groups share a common p . Estimate it as $\hat{p} = (Y_1 + Y_2)/(n_1 + n_2)$ – the **pooled proportion**.

Pooled z-statistic:

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

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⚠ Pooled vs. Unpooled SE

- **Test:** Use **pooled** $\hat{p}$ – H_0 says $p_1 = p_2$, so exploit that constraint.
- **CI:** Use **unpooled** $\hat{p}_1, \hat{p}_2$ – the CI makes no assumption about $p_1 = p_2$.

Medical Example: Adverse Event Rates

Clinical Trial: A drug trial compares 30-day SAE rates: treatment ($n_1 = 150, y_1 = 12$) vs. control ($n_2 = 148, y_2 = 24$). Test $H_0 : p_1 = p_2$ and construct a 95% CI.

```
1 n1 <- 150; y1 <- 12
2 n2 <- 148; y2 <- 24
3
4 p1_hat <- y1/n1; p2_hat <- y2/n2
5 p_pool <- (y1 + y2) / (n1 + n2)
6
7 # Test (pooled SE)
8 se_pool <- sqrt(p_pool*(1-p_pool)*(1/n1 + 1/n2))
9 z_stat <- (p1_hat - p2_hat) / se_pool
10 p_val <- 2 * pnorm(abs(z_stat), lower.tail = FALSE)
11
12 # 95% CI (unpooled SE)
13 se_ci <- sqrt(p1_hat*(1-p1_hat)/n1 + p2_hat*(1-p2_hat)/n2)
14 ci <- (p1_hat - p2_hat) + c(-1, 1) * qnorm(0.975) * se_ci
15
16 cat(sprintf("p1_hat = %.4f | p2_hat = %.4f | pooled p = %.4f\n",
17             p1_hat, p2_hat, p_pool))
```

p1_hat = 0.0800 | p2_hat = 0.1622 | pooled p = 0.1208

```
1 cat(sprintf("Z = %.3f | p-value = %.4f\n", z_stat, p_val))
```

Compare with prop.test(), and Visualize

```
1 prop.test(c(y1, y2), c(n1, n2), correct = FALSE) |> tidy()
```

```
# A tibble: 1 × 9
```

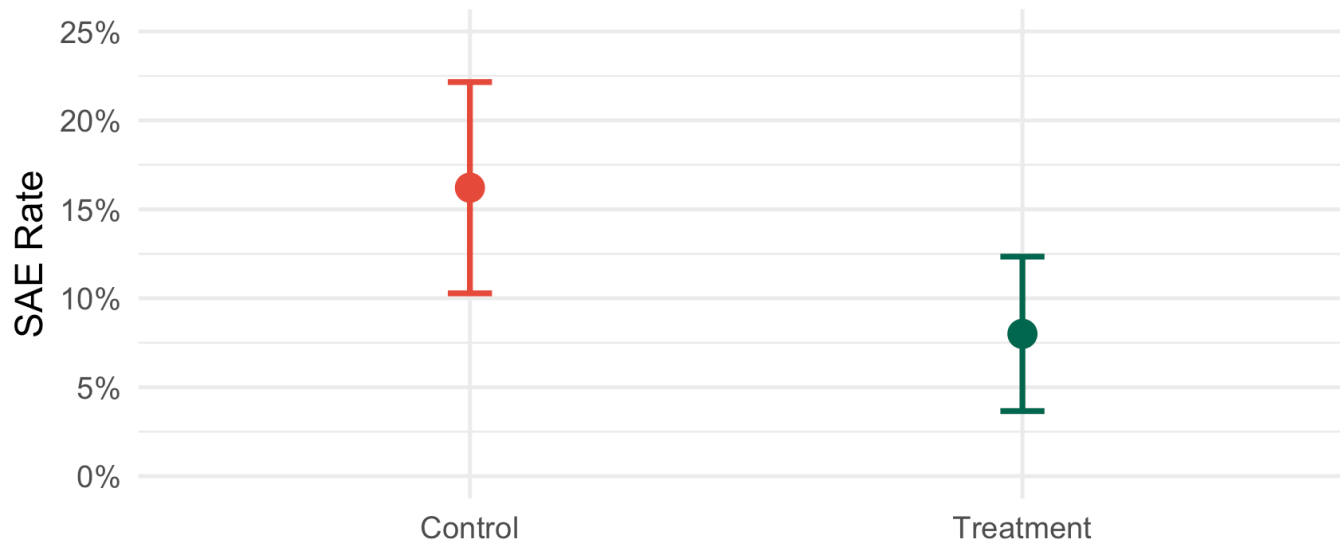
	estimate1	estimate2	statistic	p.value	parameter	conf.low	conf.high	method
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	0.08	0.162	4.73	0.0296	1	-0.156	-0.00860	2-sample t...

```
# i 1 more variable: alternative <chr>
```

► Code

30-Day SAE Rate by Treatment Arm

Points: estimated rates; bars: 95% Wald CIs



Relative Risk and Odds Ratio

The difference $p_1 - p_2$ (risk difference) is one effect measure. Medical research often reports:

$$\text{Relative Risk (RR)} = \frac{p_1}{p_2}, \quad \text{Odds Ratio (OR)} = \frac{p_1/(1 - p_1)}{p_2/(1 - p_2)}$$

CI for RR and OR are constructed on the log scale (approximately normal), then back-transformed.

```
1 RR <- p1_hat / p2_hat
2 OR <- (p1_hat/(1-p1_hat)) / (p2_hat/(1-p2_hat))
3
4 # Delta-method SE on log scale
5 se_logRR <- sqrt(1/y1 - 1/n1 + 1/y2 - 1/n2)
6 se_logOR <- sqrt(1/y1 + 1/(n1-y1) + 1/y2 + 1/(n2-y2))
7
8 cat(sprintf("RR = %.3f  (95%% CI: %.3f, %.3f)\n",
9           RR, exp(log(RR) - 1.96*se_logRR), exp(log(RR) + 1.96*se_logRR)))
```

RR = 0.493 (95% CI: 0.256, 0.949)

```
1 cat(sprintf("OR = %.3f  (95%% CI: %.3f, %.3f)\n",
2           OR, exp(log(OR) - 1.96*se_logOR), exp(log(OR) + 1.96*se_logOR)))
```

OR = 0.449 (95% CI: 0.216, 0.936)

Relative Risk and Odds Ratio

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5 se_logRR <- sqrt(1/y1 - 1/n1 + 1/y2 - 1/n2)
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7
8 cat(sprintf("RR = %.3f  (95%% CI: %.3f, %.3f)\n",
9           RR, exp(log(RR) - 1.96*se_logRR), exp(log(RR) + 1.96*se_logRR)))
```

RR = 0.493 (95% CI: 0.256, 0.949)

```
1 cat(sprintf("OR = %.3f  (95%% CI: %.3f, %.3f)\n",
2           OR, exp(log(OR) - 1.96*se_logOR), exp(log(OR) + 1.96*se_logOR)))
```

OR = 0.449 (95% CI: 0.216, 0.936)

Interpretation: The treatment arm had approximately half the SAE rate of control (RR $\approx$ 0.49). When p is small,



Inferences About Two Variances (DBC 9.6)

The F-Distribution

When comparing variability between two groups, we need the **F-distribution**.

Key result: If $U \sim \chi_m^2$ and $V \sim \chi_n^2$ are independent, then:

$$F = \frac{U/m}{V/n} \sim F_{m,n}$$

Connection to sample variances: Under normality, $(n_i - 1)S_i^2/\sigma_i^2 \sim \chi_{n_i-1}^2$ independently, so:

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{n_1-1, n_2-1}$$

Under $H_0 : \sigma_1^2 = \sigma_2^2$, the ratio simplifies to $F = S_1^2/S_2^2$.

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Under $H_0 : \sigma_1^2 = \sigma_2^2$, the ratio simplifies to $F = S_1^2/S_2^2$.

Properties of $F_{m,n}$:

- Support: $(0, \infty)$; right-skewed for small m, n
- $E[F] = n/(n - 2)$ for $n > 2$
- **Reciprocal:** if $F \sim F_{m,n}$ then $1/F \sim F_{n,m}$
- Critical values in R: `qf(p, df1, df2)`

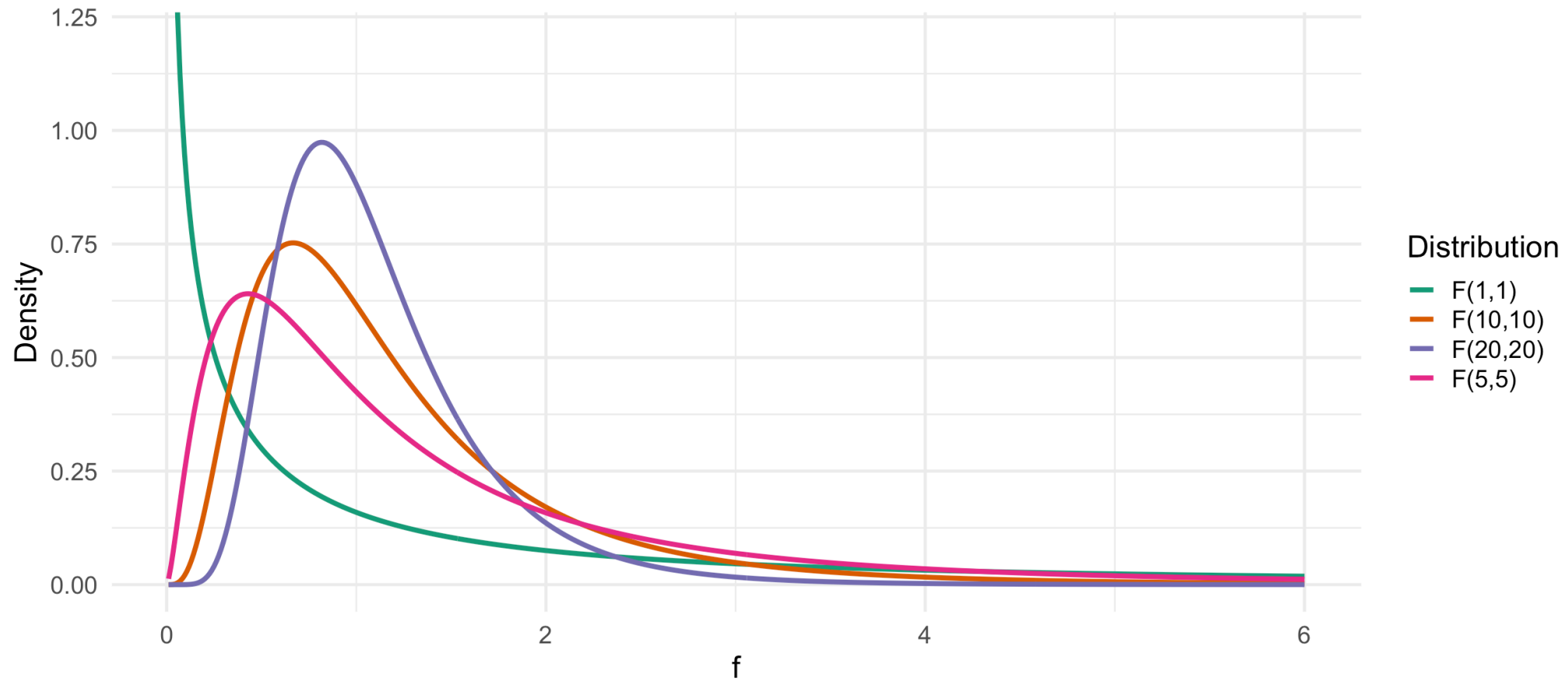


F-Distribution Shape

► Code

F-Distribution for Various Degrees of Freedom (df1 = df2)

As degrees of freedom increase, the F-distribution concentrates near 1



F-Test for $H_0 : \sigma_1^2 = \sigma_2^2$ (DBC 9.6)

Test statistic: $f = s_1^2/s_2^2$; under $H_0, F = S_1^2/S_2^2 \sim F_{n_1-1, n_2-1}$

Alternative	Rejection Region	p-value
$\sigma_1^2 > \sigma_2^2$	$f > F_{\alpha, n_1-1, n_2-1}$	$P(F \geq f)$
$\sigma_1^2 < \sigma_2^2$	$f < F_{1-\alpha, n_1-1, n_2-1}$	$P(F \leq f)$
$\sigma_1^2 \neq \sigma_2^2$	$f < F_{1-\alpha/2}$ or $f > F_{\alpha/2}$	$2 \min(P(F \leq f), P(F \geq f))$

CI for σ_1^2/σ_2^2 :

$$\left(\frac{s_1^2/s_2^2}{F_{\alpha/2, n_1-1, n_2-1}}, \frac{s_1^2/s_2^2}{F_{1-\alpha/2, n_1-1, n_2-1}} \right)$$

Note: use the reciprocal property $F_{1-\alpha/2, m, n} = 1/F_{\alpha/2, n, m}$.

Medical Example: Comparing Lab Variability

Study: Two clinical labs each measure serum potassium (mEq/L) on the same patient samples.

- Lab 1: $n_1 = 16$, $s_1 = 0.38$ mEq/L
- Lab 2: $n_2 = 21$, $s_2 = 0.22$ mEq/L

Test $H_0 : \sigma_1^2 = \sigma_2^2$ vs. $H_a : \sigma_1^2 > \sigma_2^2$ at $\alpha = 0.05$.

```
1 n1 <- 16; s1 <- 0.38
2 n2 <- 21; s2 <- 0.22
3
4 f_stat <- s1^2 / s2^2
5 df1 <- n1 - 1; df2 <- n2 - 1
6
7 p_val <- pf(f_stat, df1 = df1, df2 = df2, lower.tail = FALSE)
8
9 f_lo <- qf(0.025, df1 = df1, df2 = df2)
10 f_hi <- qf(0.975, df1 = df1, df2 = df2)
11 ci_lo <- f_stat / f_hi; ci_hi <- f_stat / f_lo
12
13 cat(sprintf("F = %.3f (df: %d, %d) | one-sided p = %.4f\n",
14             f_stat, df1, df2, p_val))
```

F = 2.983 (df: 15, 20) | one-sided p = 0.0120

F-Test in R and Sensitivity Warning

```
1 set.seed(551)
2 lab1 <- rnorm(16, mean = 4.2, sd = 0.38)
3 lab2 <- rnorm(21, mean = 4.2, sd = 0.22)
4
5 var.test(lab1, lab2, alternative = "greater") |> tidy()
```

A tibble: 1 × 9

	estimate	num.df	den.df	statistic	p.value	conf.low	conf.high	method	alternative
	<dbl>	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<chr>
1	3.04	15	20	3.04	0.0109	1.38	Inf	F tes...	greater

F-Test in R and Sensitivity Warning

```
1 set.seed(551)
2 lab1 <- rnorm(16, mean = 4.2, sd = 0.38)
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	estimate	num.df	den.df	statistic	p.value	conf.low	conf.high	method	alternative
	<dbl>	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<chr>
1	3.04	15	20	3.04	0.0109	1.38	Inf	F tes...	greater

⚠ The F-test is Extremely Sensitive to Non-Normality

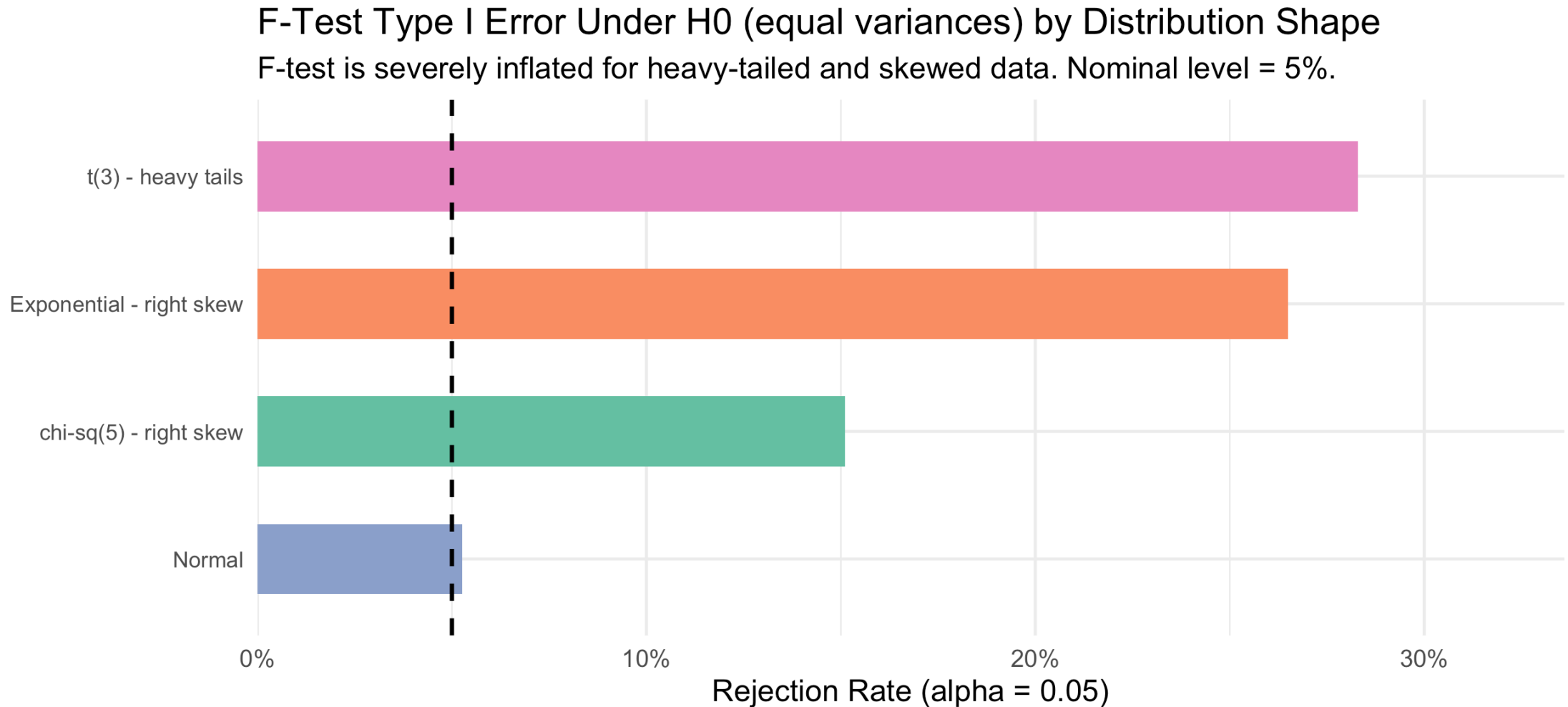
The F-test for equality of variances assumes **exact normality** of both populations — far more sensitive than the t-test for means. For non-normal data:

- **Levene's test** (`car::leveneTest()`) — uses absolute deviations from group medians; robust to non-normality
- **Bartlett's test** (`bartlett.test()`) — equivalent power to F-test but also assumes normality

In medical research, the F-test for variances is most appropriate when normality is well-established (e.g., duplicate laboratory measurements). In most other settings, Levene's test is preferred.

F-Test Sensitivity to Non-Normality: Simulation

► Code



Complete Clinical Trial Example

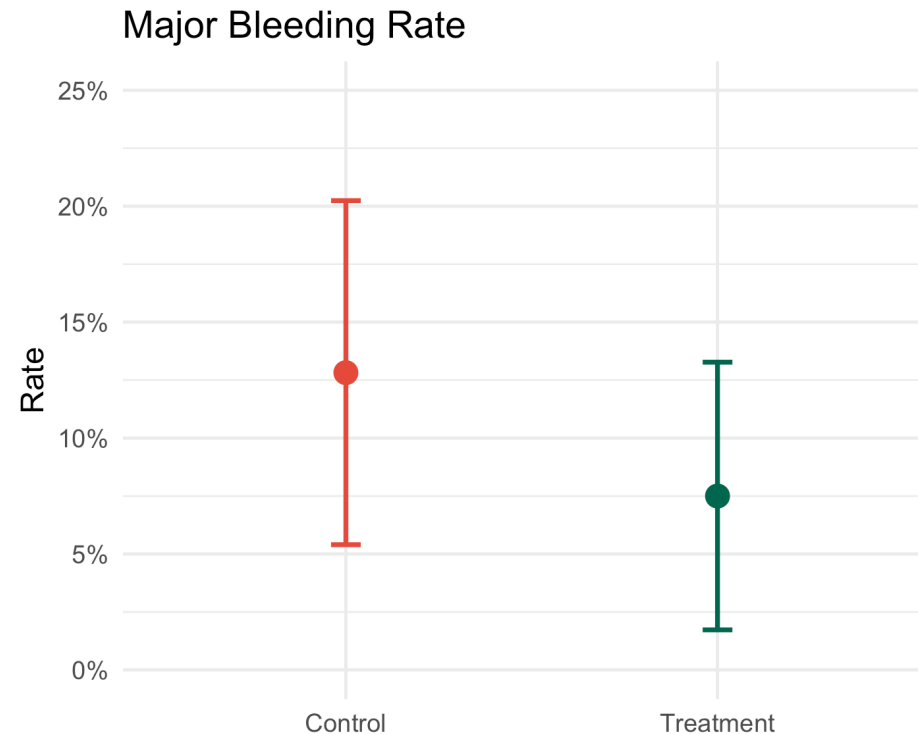
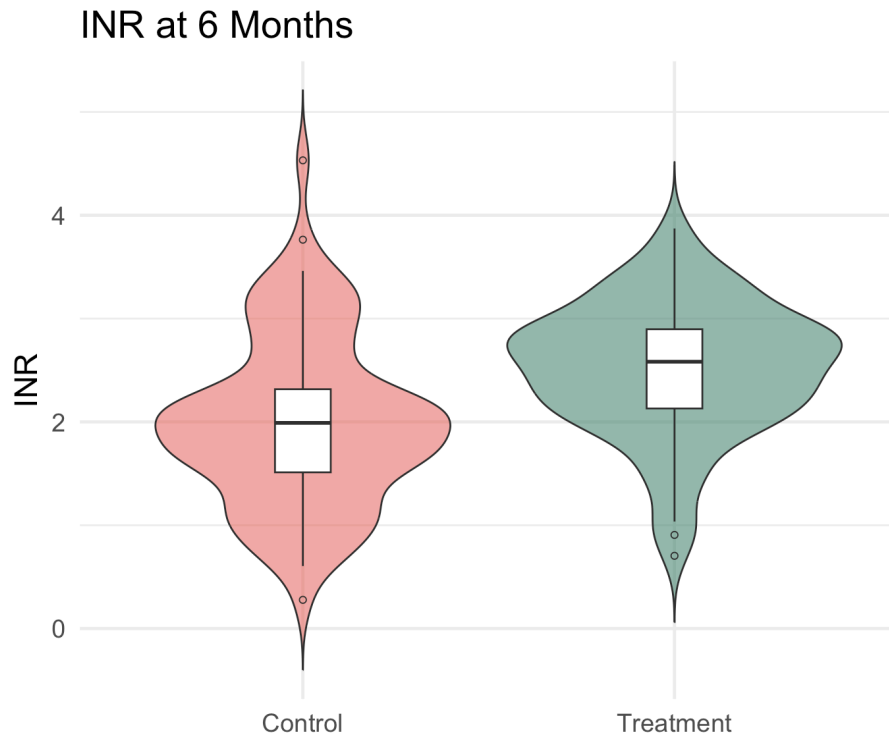
Anticoagulant Trial: Applying All Methods

RCT: New anticoagulant vs. standard of care in atrial fibrillation patients. Outcomes at 6 months: INR (continuous) and major bleeding events (binary).

► Code

Anticoagulant RCT: Two-Sample Analyses

Left: continuous outcome (Welch t-test + F-test) | Right: binary outcome (z-test for proportions)



Reading All Results Together

```
1 cat("=== INR: Welch t-test ===\n")
```

=== INR: Welch t-test ===

```
1 cat(sprintf("Treatment mean: %.2f | Control mean: %.2f\n",
2             mean(inr_treat), mean(inr_ctrl)))
```

Treatment mean: 2.51 | Control mean: 1.99

```
1 cat(sprintf("Difference: %.2f (95%% CI: %.2f, %.2f)  p = %.4f\n",
2             welch_res$estimate, welch_res$conf.low,
3             welch_res$conf.high, welch_res$p.value))
```

Difference: 0.52 (95% CI: 0.28, 0.75) p = 0.0000

```
1 cat("\n=== INR: F-test for equal variances ===\n")
```

=== INR: F-test for equal variances ===

```
1 cat(sprintf("S_treat = %.2f | S_ctrl = %.2f | F = %.3f | p = %.4f\n",
2             sd(inr_treat), sd(inr_ctrl),
3             ftest_res$statistic, ftest_res$p.value))
```

S_treat = 0.64 | S_ctrl = 0.82 | F = 0.612 | p = 0.0312

```
1 cat("\n=== Bleeding Rate: Two-sample z-test ===\n")
```

=== Bleeding Rate: Two-sample z-test ===

```
1 cat(sprintf("Treatment: %.1f%% | Control: %.1f%%\n",
```

Lesson 18 - Two-Sample Inference



Key Takeaways

Key Takeaways: Means (DBC 9.5)

! Two-Sample Tests for Means

1. **Point estimator** $\bar{X}_1 - \bar{X}_2$ is unbiased; approximately normal by CLT (both n_i large) or exact normality of populations
2. **Welch t-test** (default): $T_W \approx t_{\hat{\nu}}$; Satterthwaite df accounts for unequal variances; valid whether or not $\sigma_1^2 = \sigma_2^2$
3. **Pooled t-test**: exact under normality + $\sigma_1^2 = \sigma_2^2$; requires strong prior justification — anti-conservative when variances differ and n 's are unequal
4. **CI-test duality**: the $100(1 - \alpha)\%$ CI for Δ contains exactly the Δ_0 values for which the two-sided test fails to reject at level α

Key Takeaways: Proportions and Variances (DBC 9.6)

! Two-Sample Tests for Proportions and Variances

1. **Proportions (CLT):** $\hat{p}_i$ is a sample mean of Bernoulli observations; CLT gives $\hat{p}_1 - \hat{p}_2 \sim N(p_1 - p_2, SE^2)$; requires $n_i \hat{p}_i \geq 5$ and $n_i(1 - \hat{p}_i) \geq 5$
2. **Pooled vs. unpooled SE:** Use **pooled** $\hat{p}$ for the test (exploits $H_0 : p_1 = p_2$); use **unpooled** for the CI (no such assumption)
3. **F-test:** $S_1^2/S_2^2 \sim F_{n_1-1, n_2-1}$ under normality and $H_0 : \sigma_1^2 = \sigma_2^2$; the CI for σ_1^2/σ_2^2 is asymmetric
4. **Sensitivity:** The F-test for variances is *extremely* sensitive to non-normality — prefer Levene's test in practice for most biomedical data
5. **Always report** point estimates + CIs; choose the effect measure (RD, RR, OR) that is most interpretable in context

Looking Ahead

Next class: Analysis of Paired Data (DBC 10.3)

- When to use a paired design (pre/post, crossover trials, matched pairs)
- Reduction to one-sample problem: $D_i = X_{1i} - X_{2i}$
- Paired t-test, Wilcoxon signed-rank test, sign test

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- Reduction to one-sample problem: $D_i = X_{1i} - X_{2i}$
- Paired t-test, Wilcoxon signed-rank test, sign test

Lesson 20: Bootstrap and permutation methods for two samples; course review

References

- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Sections 9.5–9.6.
- Welch, B. L. (1947). The generalization of 'Student's' problem when several different population variances are involved. *Biometrika*, 34(1–2), 28–35.
- Chihara, L. M. and Hesterberg, T. C. *Mathematical Statistics with Resampling and R* (Wiley). Chapter 3.

Questions?

Thank you!